K L HYDERABAD

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Sports Registration Application

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UNDER THE ESTEEMED GUIDANCE OF

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CERTIFICATE

This is to certify that the project based laboratory report entitled **Sports Registration Application** submitted by **B. Charan Teja , G. Channa Reddy** bearing Regd. No. **2310040035,2310040083,** to the **Department of Basic Engineering Sciences, KL University** in partial fulfillment of the requirements for the completion of a project in **"Computational Thinking for Object Oriented Design -23SC1203**" course in I Year B Tech II Semester, is a Bonafide record of the work carried out by him under my supervision during the academic year 2023-24.

PROJECT SUPERVISOR **HEAD OF THE DEPARTMENT**

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ABSTRACT

The Sports Registration Application is a user-friendly graphical interface designed to simplify the process of registering for sports events in an educational institution. Developed using Java Swing, the application allows students to input their personal details including name, year, course, branch, and date of birth. The course selection dynamically updates available branches, offering tailored options such as B Tech (ECE, CSE, AI&DS) or Degree (BBA, BCA). Stringent validation ensures data integrity, prohibiting numerical characters in names and enforcing a dd/mm/yyyy format for dates of birth. Students can register for individual or double participation in sports like Basketball, Volleyball, Badminton, Tennis, and Table Tennis. Upon successful registration, the application generates a comprehensive confirmation message, incorporating all entered details alongside the selected sport and participation type. This streamlined process eliminates paperwork and enhances efficiency, facilitating seamless participation in sports activities within the educational institution.

INDEX

S.NO	TITLE	PAGE NO
1	Introduction	6
2	Aim of the Project	7
2.1	Advantages & Disadvantages	7
2.2	Future Implementation	7
3	Software & Hardware Details	8
4	Class Diagram	9
5	Implementation	10
6	Outputs/ScreenShots	14
7	Conclusion & Future Scope	19

INTRODUCTION

In today's world, managing sports registrations can be a time-consuming and cumbersome process, often relying on paper forms and manual data entry. This project aims to develop a user-friendly sports registration application that streamlines the process for both participants and organizers.

The application offers a convenient way to:

1. **Simplify Registration:** Reduce the need for paper forms and manual data entry for participants.
2. **Improve Data Management**: Organize participant information in a centralized and easily accessible format for organizers.
3. **Enhance User Experience**: Provide a clear and intuitive interface for both participants registering and organizers managing registrations.
4. **Increase Efficiency:** Streamline the registration process, saving time and resources for all parties involved.

This project focuses on developing a user-friendly application with the following key features:

1. **Intuitive Interface:** Clear labels and instructions for participants and organizers.
2. **Flexible Registration Options:** Support for individual and double participation types.
3. **Dynamic Branch Selection**: Automatic adjustment of branch options based on chosen course.
4. **Customizable Validation**: Validation logic to ensure accurate data entry (details to be specified later).
5. **Data Persistence**: Secure saving of registration information to a CSV file for easy access and record-keeping.

AIM

The primary aim of the Sports Registration App project is to develop an efficient, user-friendly, and reliable software application that streamlines the process of registering participants for various sports events. This project seeks to address the challenges associated with traditional paper-based registration methods by providing a digital solution that enhances accuracy, reduces administrative overhead, and improves the overall user experience for both participants and event organizers.

Advantages: -

- **Saves Time:** Streamlines registration for participants and organizers, reducing paperwork and manual data entry.
- **Improved Organization:** Centralizes participant information for easy access and management.
- **Flexibility:** Supports various participation types and can be customized with validation rules.
- **Cost-Effective:** Reduces printing and administrative costs associated with paper forms.

Disadvantages: -

- **Technology Dependence:** Requires internet access and might exclude those with limited technological resources.
- **Learning Curve:** Users might need time to learn the application's interface.
- **Data Security:** Requires security measures to protect participant information.
- **Limited Features (Optional):** Initial version might have limited functionalities compared to established platforms.
- **Maintenance:** Needs ongoing maintenance and updates for continued functionality and security.

SYSTEM REQUIREMENTS

- **SOFTWARE REQUIREMENTS:**

The major software requirements of the project are as follows:

Language	:	JAVA
Operating system	:	Windows 10 Or Later.
Tools	:	Eclipse IDE

- **HARDWARE REQUIREMENTS:**

The hardware requirements that map towards the software are as follows:

RAM	:	8GB or More
Processor	:	Intel 32/64 Bit with 1GHZ or More Speed

CLASS DIAGRAMS

IMPLEMENTATION

```java
package Exam;

import javax.swing.*;
import java.awt.event.*;
import java.io.FileWriter;
import java.io.IOException;

public class SportsRegistrationApp1 extends JFrame implements ActionListener {

    private JTextField nameField;
    private JComboBox<String> yearComboBox;
    private JComboBox<String> courseComboBox;
    private JComboBox<String> branchComboBox;
    private JTextField dobField;
    private JTextField name2Field;
    private JTextField dob2Field;
    private JComboBox<String> sportComboBox;
    private JComboBox<String> participationComboBox;
    private JButton registerButton;

    private String[] years = {"1", "2", "3", "4"};
    private String[] courses = {"B Tech", "Degree"};
    private String[] bTechBranches = {"ECE", "CSE", "AI&DS"};
    private String[] degreeBranches = {"BBA", "BCA"};

    private String[] sports = {"Basketball", "Volleyball", "Badminton", "Tennis", "Table Tennis"};
    private String[] participationTypes = {"Single", "Double"};

    public SportsRegistrationApp1() {
        setTitle("Sports Registration App");
        setSize(400, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(null);

        nameField = new JTextField();
        nameField.setBounds(120, 20, 200, 25);
        add(nameField);

        yearComboBox = new JComboBox<>(years);
        yearComboBox.setBounds(120, 50, 100, 25);
        add(yearComboBox);

        courseComboBox = new JComboBox<>(courses);
        courseComboBox.setBounds(120, 80, 100, 25);
        add(courseComboBox);

        courseComboBox.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                String selectedCourse = (String) courseComboBox.getSelectedItem();
                if (selectedCourse.equals("B Tech")) {
                    branchComboBox.setModel(new DefaultComboBoxModel<>(bTechBranches));
                } else if (selectedCourse.equals("Degree")) {
                    branchComboBox.setModel(new DefaultComboBoxModel<>(degreeBranches));
                }
            }
        });
```

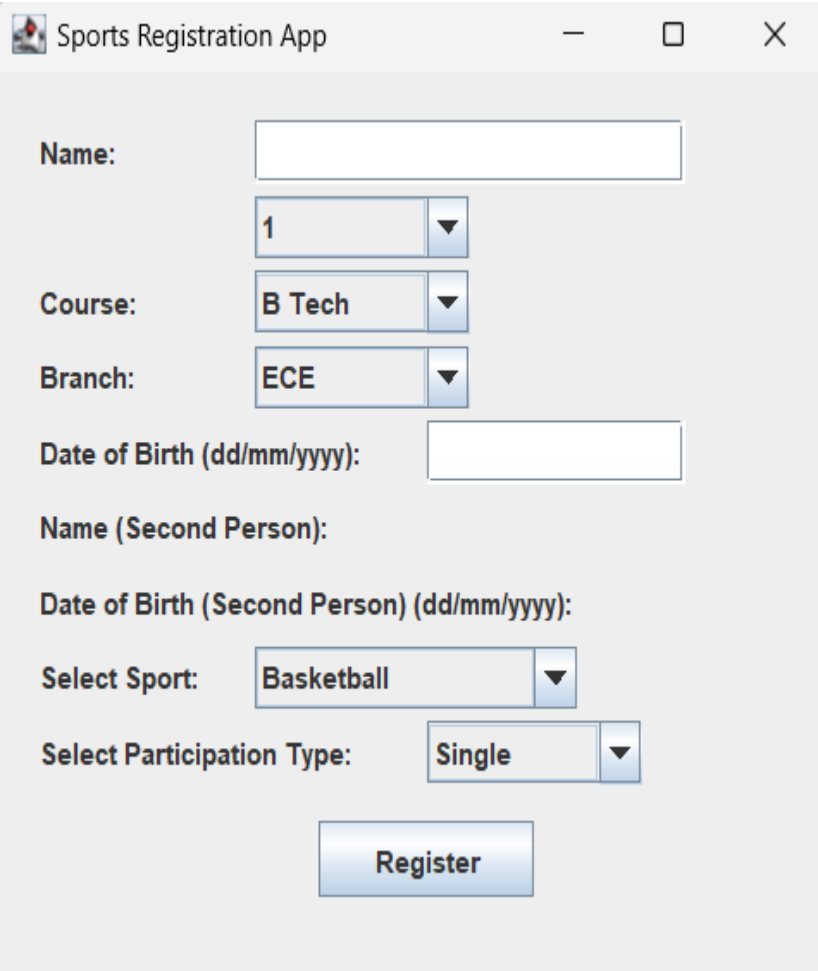
```
branchComboBox = new JComboBox<>(bTechBranches);
branchComboBox.setBounds(120, 110, 100, 25);
add(branchComboBox);

dobField = new JTextField();
dobField.setBounds(200, 140, 120, 25);
add(dobField);

name2Field = new JTextField();
name2Field.setBounds(200, 170, 120, 25);
add(name2Field);
name2Field.setVisible(false);

dob2Field = new JTextField();
dob2Field.setBounds(200, 200, 120, 25);
add(dob2Field);
dob2Field.setVisible(false);

sportComboBox = new JComboBox<>(sports);
sportComboBox.setBounds(120, 230, 150, 25);
add(sportComboBox);

participationComboBox = new JComboBox<>(participationTypes);
participationComboBox.setBounds(200, 260, 100, 25);
add(participationComboBox);

participationComboBox.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String selected = (String) participationComboBox.getSelectedItem();
        if (selected.equals("Double")) {
            name2Field.setVisible(true);
            dob2Field.setVisible(true);
        } else {
            name2Field.setVisible(false);
            dob2Field.setVisible(false);
        }
    }
});

registerButton = new JButton("Register");
registerButton.setBounds(150, 300, 100, 30);
registerButton.addActionListener(this);
add(registerButton);

JLabel nameLabel = new JLabel("Name:");
nameLabel.setBounds(20, 20, 80, 25);
add(nameLabel);

JLabel yearLabel = new JLabel("Year:");
yearLabel.setBounds(20, 50, 80, 25);
add(yearLabel);

JLabel courseLabel = new JLabel("Course:");
courseLabel.setBounds(20, 80, 80, 25);
add(courseLabel);

JLabel branchLabel = new JLabel("Branch:");
branchLabel.setBounds(20, 110, 80, 25);
add(branchLabel);
```

```
        JLabel dobLabel = new JLabel("Date of Birth (dd/mm/yyyy):");
        dobLabel.setBounds(20, 140, 180, 25);
        add(dobLabel);

        JLabel nameLabel2 = new JLabel("Name (Second Person):");
        nameLabel2.setBounds(20, 170, 150, 25);
        add(nameLabel2);

        JLabel dobLabel2 = new JLabel("Date of Birth (Second Person) (dd/mm/yyyy):");
        dobLabel2.setBounds(20, 200, 250, 25);
        add(dobLabel2);

        JLabel sportLabel = new JLabel("Select Sport:");
        sportLabel.setBounds(20, 230, 100, 25);
        add(sportLabel);

        JLabel participationLabel = new JLabel("Select Participation Type:");
        participationLabel.setBounds(20, 260, 180, 25);
        add(participationLabel);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == registerButton) {
            if (!validateInputs()) {
                JOptionPane.showMessageDialog(this, "Please fill in all fields correctly.", "Error",
JOptionPane.ERROR_MESSAGE);
                return;
            }

            String name = nameField.getText();
            String year = (String) yearComboBox.getSelectedItem();
            String course = (String) courseComboBox.getSelectedItem();
            String branch = (String) branchComboBox.getSelectedItem();
            String dob = dobField.getText();
            String sport = (String) sportComboBox.getSelectedItem();
            String participationType = (String) participationComboBox.getSelectedItem();

            String message = "Name: " + name + "\nYear: " + year + "\nCourse: " + course + "\nBranch: " + branch
+
                    "\nDate of Birth: " + dob + "\nRegistered Sport: " + sport + " (" + participationType +
")\n\nRegistration Successful!";
            if (participationType.equals("Double")) {
                String name2 = name2Field.getText();
                String dob2 = dob2Field.getText();
                message += "\nSecond Person Name: " + name2 + "\nSecond Person Date of Birth: " + dob2;
            }

            saveToCSV(name, year, course, branch, dob, sport, participationType,
participationType.equals("Double") ? name2Field.getText() : "", participationType.equals("Double") ?
dob2Field.getText() : "");

            JOptionPane.showMessageDialog(this, message);
        }
    }

    private boolean validateInputs() {
        String name = nameField.getText();
        String dob = dobField.getText();
        String participationType = (String) participationComboBox.getSelectedItem();
```

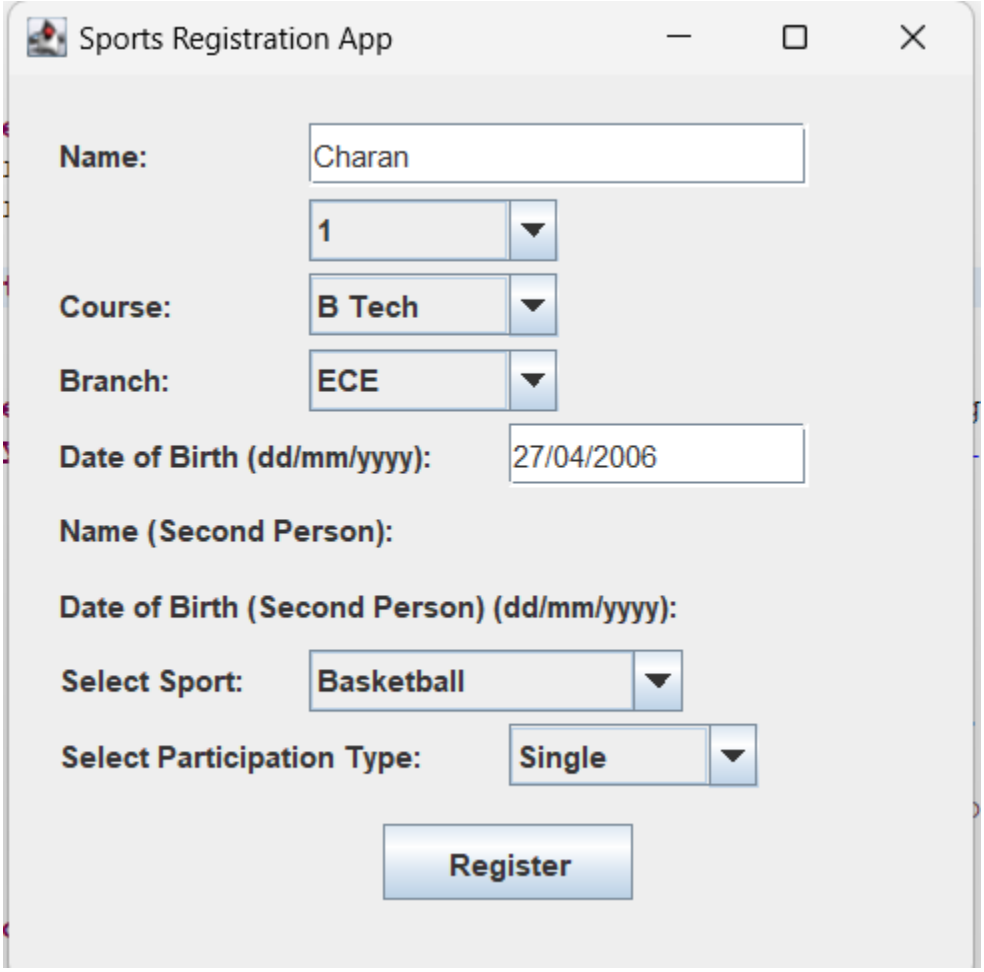
```
        if (name.isEmpty() || dob.isEmpty() || !dob.matches("\\d{2}/\\d{2}/\\d{4}")) {
            return false;
        }

        if (participationType.equals("Double")) {
            String name2 = name2Field.getText();
            String dob2 = dob2Field.getText();
            if (name2.isEmpty() || dob2.isEmpty() || !dob2.matches("\\d{2}/\\d{2}/\\d{4}")) {
                return false;
            }
        }

        return true;
    }

    private void saveToCSV(String name, String year, String course, String branch, String dob, String sport,
String participationType, String name2, String dob2) {
        try (FileWriter writer = new FileWriter("registration_data.csv", true)) {
            writer.append(name).append(',')
                    .append(year).append(',')
                    .append(course).append(',')
                    .append(branch).append(',')
                    .append(dob).append(',')
                    .append(sport).append(',')
                    .append(participationType).append(',');
            if (participationType.equals("Double")) {
                writer.append(name2).append(',').append(dob2);
            }
            writer.append('\n');
        } catch (IOException ex) {
            ex.printStackTrace();
            JOptionPane.showMessageDialog(this, "Error saving data: " + ex.getMessage(), "Error",
JOptionPane.ERROR_MESSAGE);
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(new Runnable() {
            public void run() {
                new SportsRegistrationApp1().setVisible(true);
            }
        });
    }
}
```

OUTPUTS

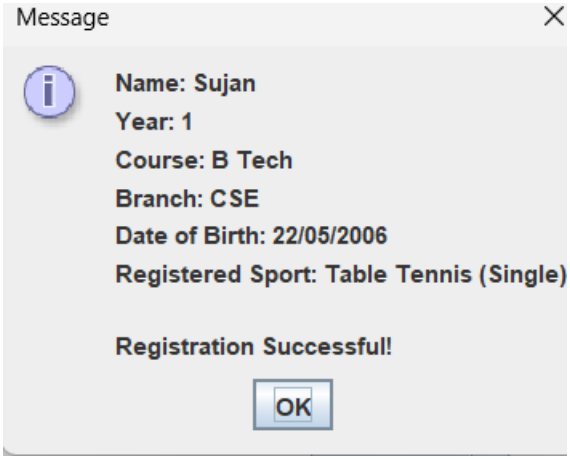
The application consists of the following details.



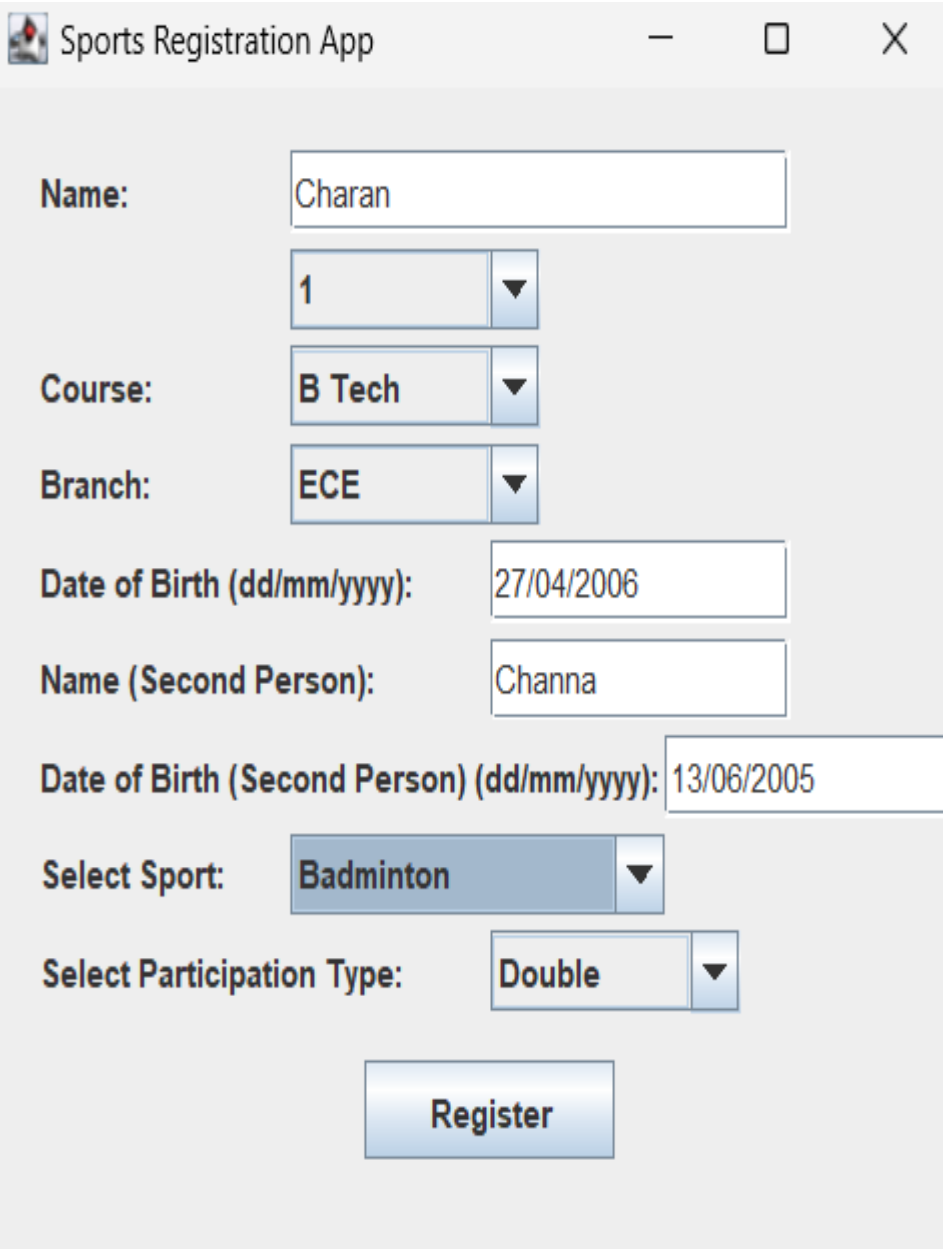
We must enter the details has asked.



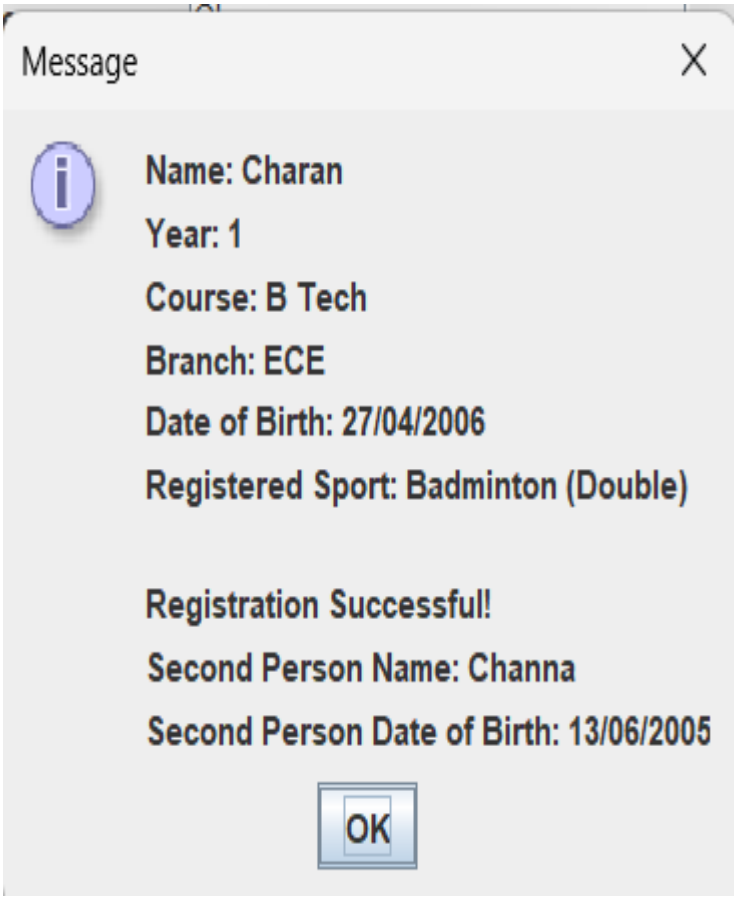
After clicking on the register, it will show a registration successful message like below.



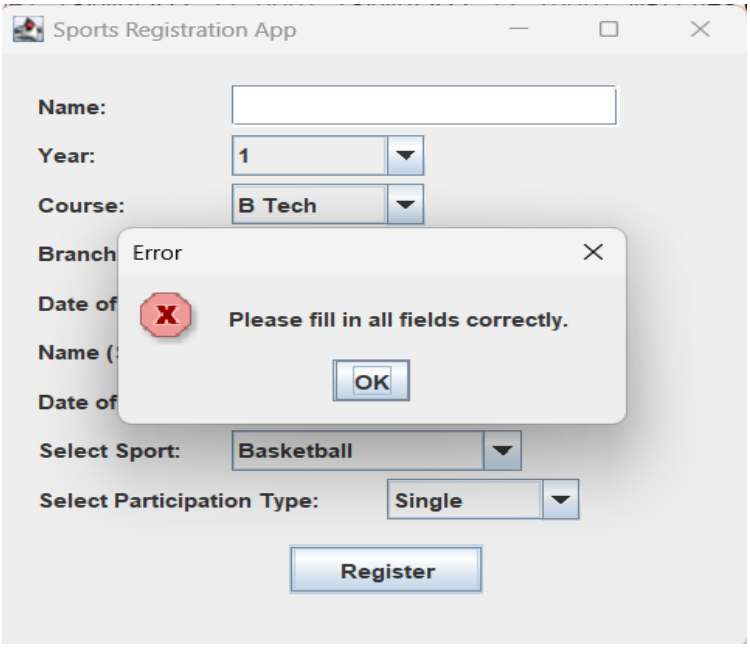
We have feature to enter the details of second person also.



After the second person registration the output message showed has below



If enter the details in incorrect manner or any details not entered, It will raise an error message.



At last, all the entered details stored in csv manner in excel are shown below.

anush	1	B Tech	ECE	30-12-2023	Basketball	Single		
Charan	1	B Tech	ECE	27-04-2006	Badmintor	Double	Channa	13-06-2005
Charan	1	B Tech	ECE		Badmintor	Double	Channa	13-06-2005
Sujan	1	B Tech	CSE	22-05-2006	Table Tenn	Single		
Nani	1	B Tech	ECE	04-07-2006	Volleyball	Single		
Manvi	1	B Tech	AI&DS	09-08-2005	Tennis	Single		
Sandya	1	B Tech	CSE	26-12-2006	Tennis	Double	Harini	31-01-2005
Santosh	1	B Tech	ECE	12-12-2005	Badmintor	Single		
Pranay	1	B Tech	CSE	12-03-2006	Table Tenn	Single		
KARTHIK	1	B Tech	CSE	15-04-2005	Badmintor	Double	SAI SUJAN	22-05-2006

CONCLUSION

In conclusion, the Sports Registration App stands as a powerful example of how thoughtful application of technology can revolutionize traditional processes. By improving efficiency, reducing errors, enhancing user experience, and providing valuable data insights, the app demonstrates the far-reaching benefits of modern software solutions. Its success serves as an inspiration for the development of similar applications across various sectors, highlighting the endless possibilities that arise from embracing digital innovation.

Overall, the Sports Registration App is a testament to the power of modern software development in enhancing operational efficiency and user experience. Its successful deployment can serve as a model for similar applications in other domains, demonstrating the potential of well-designed software solutions to transform and improve traditional processes. Whether applied in education, corporate environments, or community events, the principles and benefits exemplified by the Sports Registration App highlight the transformative impact of leveraging technology to streamline operations and improve user satisfaction.